CUSTOMER SHOPPING BEHAVIOR AND RETAIL ANALYSIS

Project Overview

The objective of this project was to analyze retail data to understand customer purchasing patterns, demographic segments, and the impact of loyalty (subscriptions). By leveraging data across different platforms, we aimed to identify high-value customer segments and optimize marketing strategies.

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Dataset summary

The project uses a shopping dataset with information on 3,900 customers. Each customer has about 19 details recorded about their habits and background.

- Key Metrics: They include Customer ID, Age, Gender, Item Purchased, Category, Purchase Amount (USD), Location, Size, Color, Season, Review Rating, and Subscription Status.
- Data Quality: We "cleaned" the data to fix any holes. For example, if a customer didn't leave a Review Rating, we filled it in with the typical rating for that category of item so the math would stay accurate.
- Ready for Analysis: The final list was organized so it could be easily plugged into professional tools like SQL and Power BI.

1. Exploratory Data Analysis using Python

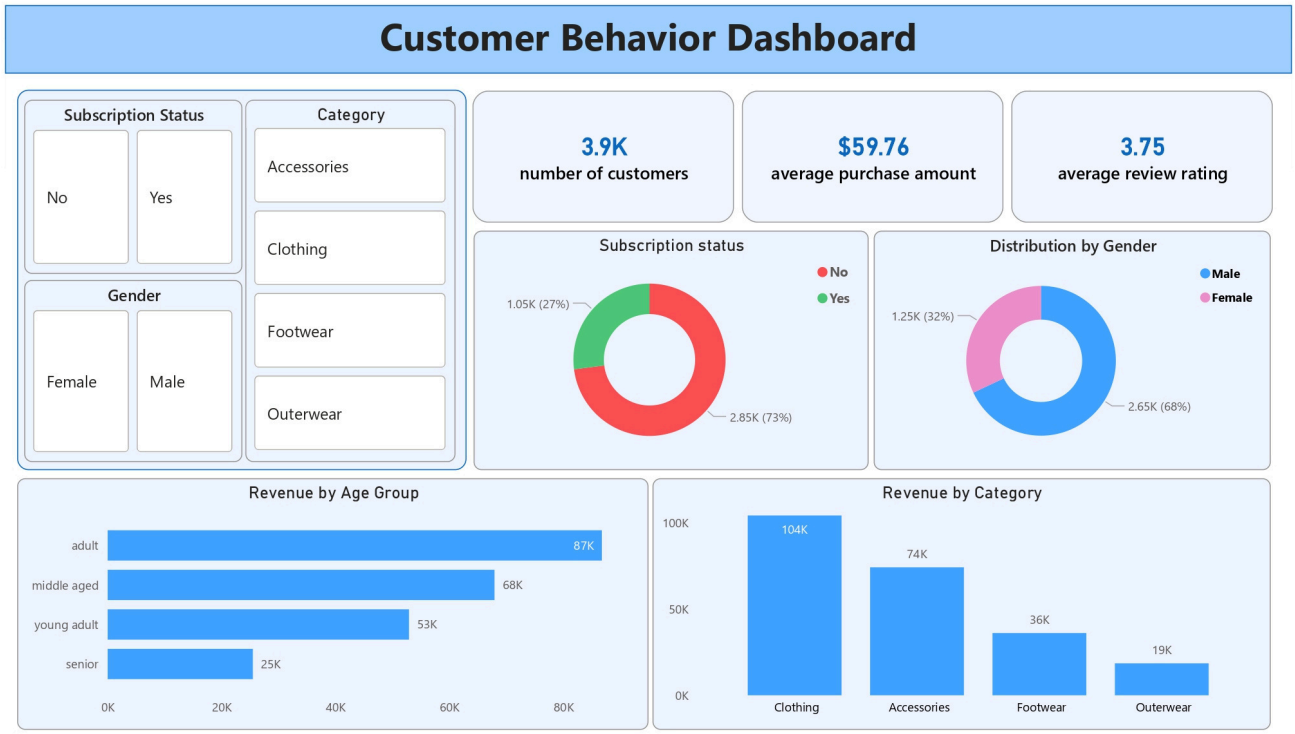
1. Loading & Inspection: After loading the CSV, **df.info()** and **df.describe()** were used to profile the 3,900 records, identifying data types and distributions across all numerical and categorical fields (age or gender).
2. Data Cleaning: For the Review Rating column, missing values are filled using the median rating of their specific Category. All column names are converted to lowercase and spaces are replaced with underscores.
3. Feature Engineering: New variables were created like Age was binned into four groups for demographic targeting, while qualitative frequency labels were mapped to numerical day values (e.g., 7 , 14 or 365).
4. Redundancy Check: By running a cross-tabulation between **discount_applied** and **promo_code_used**, it was determined that one of these columns was redundant. Consequently, **promo_code_used** was dropped
5. Database Export: The final stage of the notebook moves the data out of the local Python environment. It sets up a connection to a MySQL database using **sqlalchemy** and **pymysql**.

2. Advanced Data Analysis using SQL

1. Revenue by Gender: **Male** customers generated **$157,890** in total revenue, while **female** customers contributed $75,191.
2. High-Spending Discount Users: Several customers used a discount but still spent above the average purchase amount, including Customer ID 4 ($90), Customer ID 9 ($97), and Customer ID 20 ($90).
3. Top 5 Product Ratings: The highest-rated products are Shorts (4.9), Handbags (4.8), Dresses (4.7), Coats (4.7), and Skirts (4.6).
4. Shipping Cost Comparison: Average purchase amounts were nearly identical across shipping types, with Standard shipping at **$59.45** and Express shipping at **$59.49**.
5. Subscription Spend Impact: Subscribed customers had a higher average spend **($59.38)** compared to non-subscribers **($59.03)**, though non-subscribers accounted for higher total revenue **($154,847 vs. $78,234)** due to a larger customer base.

6. Highest Discount Rates: The products most frequently purchased with a discount are Sweaters (23.5%), Jackets (22.8%), Pants (22.2%), Shirts (21.9%), and Dresses (21.5%).
7. Customer Segmentation: The customer base consists primarily of "Returning" customers (2,500), followed by "Loyal" customers (1,000), and "New" customers (400).
8. Top Category Products: In the "Clothing" category, the top three items are Blouses, Sweaters, and Jeans; in "Footwear," they are Sandals, Sneakers, and Shoes.
9. Repeat Buyer Subscriptions: Among repeat buyers with more than 5 previous purchases, **26.5%** are currently subscribed, while **73.5%** are not.
10. Revenue by Age Group: The "Adult" group (30-49) generated the most revenue at **$88,540**, followed by "Middle Aged" (50-64) at **$65,120**, "Young Adult" (0-29) at **$52,340**, and "Senior" (65+) at **$27,081**.

3. Interactive Dashboard in Power BI



Executive Summary:

The Customer Behavior Dashboard provides a comprehensive overview of the company's current market standing, tracking a **total of 3,900 customers.** Key performance indicators reveal an **average purchase amount of $59.76** and an **average review rating of 3.75**, indicating a stable customer experience with opportunities for growth in service satisfaction

Revenue Performance and Product Mix:

- Top Performing Category: **Clothing** is the leading revenue driver, contributing **$104K**.
- Secondary Categories: Accessories follow at $74K, while Footwear ($36K) and Outerwear ($19K) represent smaller portions of the total sales mix.
- Primary Consumer Base: The "**Adult**" age group is the most profitable segment, accounting for **$87K** in revenue.
- Growth Segments: "Middle-aged" ($68K) and "Young Adult" ($53K) segments show significant engagement, while the "Senior" group contributes $25K.

Customer Segmentation and Behavior:

- Gender Distribution: The customer base is predominantly **Male (68%)**, with **Females (32%)**.
- Subscription Status: There is a significant opportunity for retention growth, as **73% of customers (2.85K)** are currently not subscribers. Only **27% (1.05K)** have an active subscription.

Interactive Functionality:

- Slicers: Users can isolate data by Subscription Status (Yes/No) or specific Categories (Accessories, Clothing, Footwear, Outerwear) to observe how metrics shift for different segments.
- Cross-Filtering: Selecting any data point—such as a specific age group or gender—will automatically update all other visuals to reflect that specific subset of the population.

4. Recommendations and Conclusions

1. Expand Subscription Incentives: Only **27%** of customers are currently subscribers. Given that "Loyal" customers drive consistent revenue, creating a bridge for "Returning" customers to subscribe is critical.
2. Targeted Demographic Campaigns: Marketing should focus on the **30-64 age segment**, as they represent the highest lifetime value.
3. Category-Specific Strategy: Clothing is the clear leader; however, "Outerwear" has room for growth. Seasonal promotions in late summer could boost winter-specific item sales.